ADC + DAC - Shafayat, Himaddri

ECL - Jishnu, Himaddri

Signal Generator + Schmitt trigger - Obaidur, Jahin

TTL + MOSFET - Farhan bhai, Afia apu

Signal Generator + Schmitt trigger [JAA]

Set-A

The figure on the left is a square wave generator constructed with a schmitt trigger. The generator has a waveform depicted by the right-handside figure.

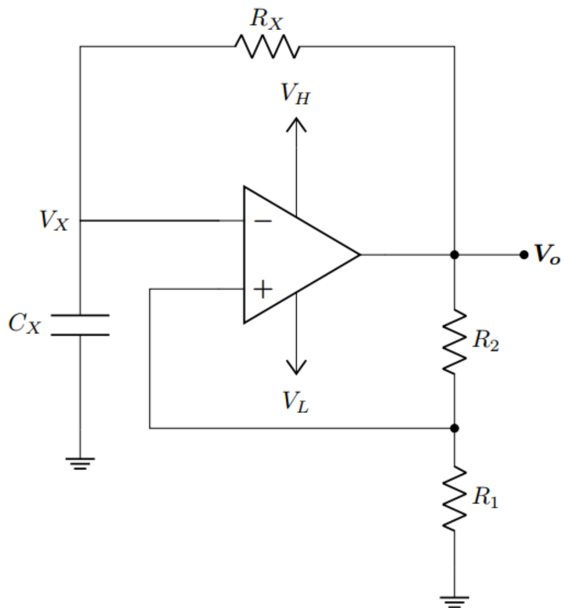


Figure 1

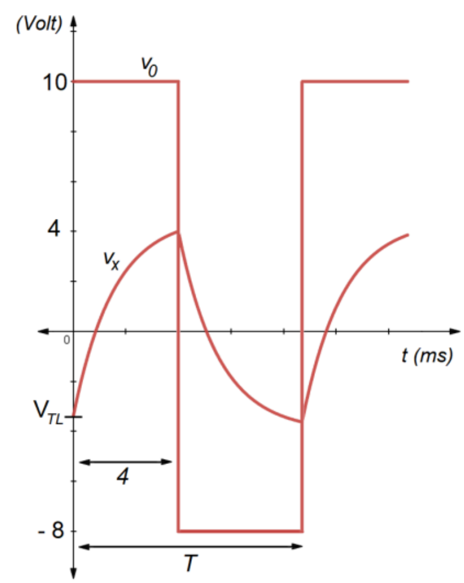


Figure 2

CO3	(a)	**Identity** whether the Schmitt Trigger in figure 1 is **inverting or non-inverting**.	[2]
CO3	(b)	**Find** the **Hysteresis Width** of the used Schmitt trigger.	[6]
CO3	(c)	Using $R_1 = 12\ k\Omega$, **show that** R_1 is smaller than R_2.	[4]
CO3	(d)	**Determine** the **duty cycle** of the square wave generator circuit. (*hint: R_x and C_x values are not required for this part*)	[8]

Solution - Set A

(a) *Inverting Schmitt Trigger*

b

$$4 = 10\,\frac{R_1}{R_1+R_2} \rightarrow \frac{V_{TL}}{4} = -8/10$$

$$V_{TL} = -8\,\frac{R_1}{R_1+R_2}$$

$$\Rightarrow V_{TL} = -32/10$$

$$= -3.2\,V$$

∴ *Hysteresis Width =* $V_{TH} - V_{TL}$ = **4 + 3.2 = 7.2 V**

c

$$4 = 10\,\frac{R_1}{R_1+R_2} \Rightarrow 4R_1 + 4R_2 = 10R_1$$

$$\Rightarrow 6R_1 = 4R_2 \Rightarrow R_2 = \frac{3}{2}R_1$$

$$\Rightarrow R_2 = \frac{3}{2} \times 12\,k\Omega = 18\,k\Omega$$

d

Find T : $T = T_1 + T_2$

$$T_1 = \tau \ln\left|\frac{10+3.2}{10-4}\right| = 4$$

$$\Rightarrow \tau = 5.073 \times 10^{-3}\ \text{sec}$$

$$\therefore T_2 = \tau \ln\left|\frac{-8-4}{-8+3.2}\right| = 4.648\,\text{ms}$$

$$\therefore T = 8.648\ \text{ms}$$

$$\text{Duty} = \frac{400}{8.648}\,\% = 46.253\%$$

Formulae :

$$T_2 = \tau \ln\left|\frac{V_L - V_{TH}}{V_L - V_{TL}}\right|$$

$$T_1 = \tau \ln\left|\frac{V_H - V_{TL}}{V_H - V_{TH}}\right|$$

$$V_{TH} = V_H\,\frac{R_1}{R_1+R_2}$$

$$V_{TL} = V_L\,\frac{R_1}{R_1+R_2}$$

Set-B

The figure on the left is a square wave generator constructed with a schmitt trigger. The generator has a waveform depicted by the right-handside figure.

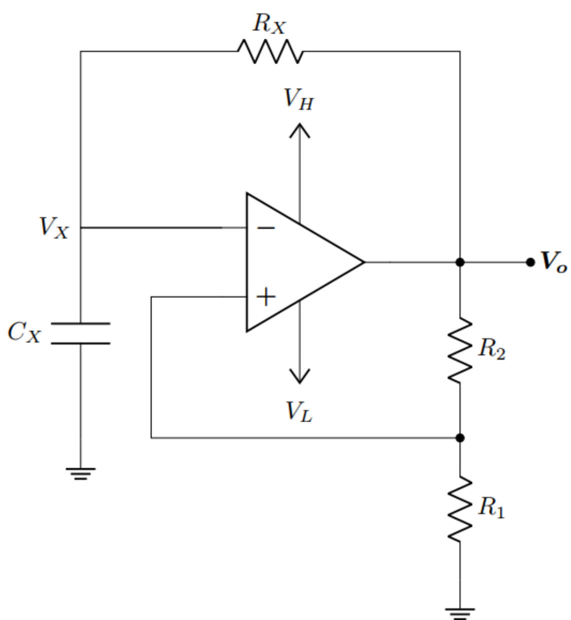


Figure 2

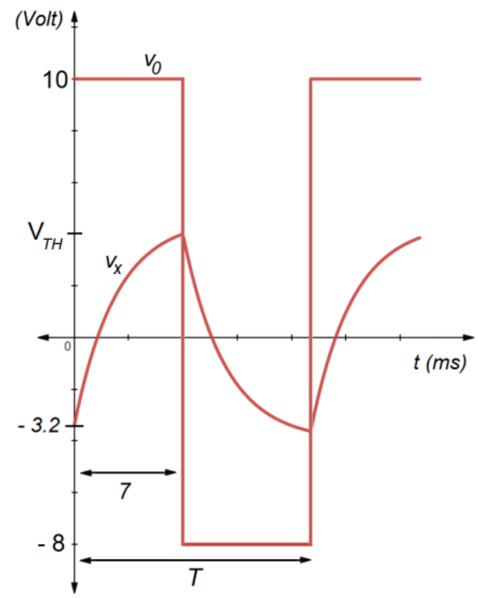


Figure 3

CO3	(a)	**Identity** whether the Schmitt Trigger in figure 2 is **inverting or non-inverting**.	[2]
CO3	(b)	**Find** the **Hysteresis Width** of the used Schmitt trigger.	[6]
CO3	(c)	Using $R_2 = 27\ k\Omega$, **show that** R_2 is higher than R_1.	[4]
CO3	(d)	**Determine** the **duty cycle** of the square wave generator circuit. (*hint: R_x and C_x values are not required for this part*)	[8]

Solution - Set B

(a) *Inverting Schmitt Trigger*

(b) $V_{TH} = (10/(-8)) * (-3.2) = -4V$. *Hysteresis = 7.2 V*

(c) $R_1 = (2/3) * 27 = 18\, k\, ohm.$ $R_2 > R_1$

(d) *Same process as set-A.*

The time constant, τ = 8.878 ms

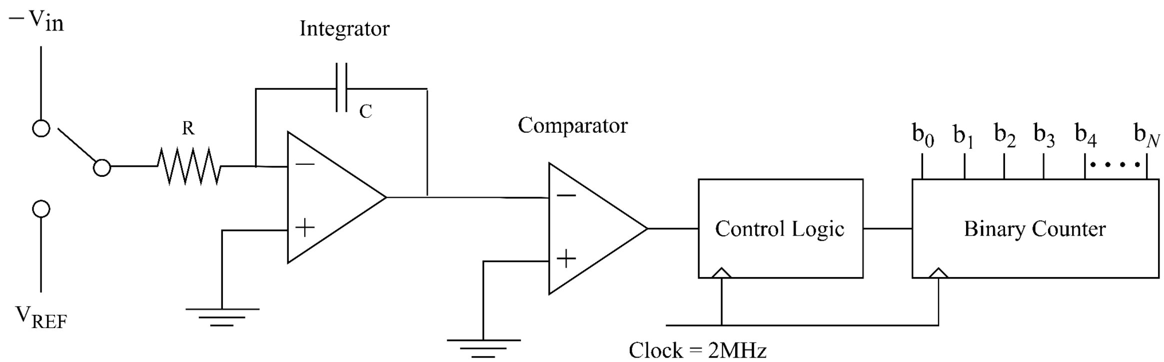
T_2 = *8.135 ms*

T = *15.135 ms*

Duty cycle = 700% / 15.135 = 46.25% (same as Set-A)

ADC + DAC [QSH]

Solution - Set A



A dual slope A/D converter has $V_{REF} = 5.12$ V, and a N bit counter that outputs N-bit representation of the input signal. A 2 MHz clock is used for the clock and counter circuits

CO2	**(a)**	**Determine** the input voltage range. If step size is 0.02 V, **determine** the total number of steps and the number of bits used in this ADC.	**7**
CO2	**(b)**	For input voltage, $V_{in} = 3$ V, **determine** the ADC output count. **Calculate** the time required to get this output.	**7**
CO2	**(c)**	**Draw Vo vs time graph** for the input voltage V_{in} mentioned in (b). If $V_{in} = 4$V, how the graph will change? Explain briefly	**6**

Solution:

(a) Input Voltage range = (V_{REF} - 0) = 5.12 V

$$\text{Total Number of Steps} = \frac{Voltage\ range}{step\ size}$$

$$= \frac{5.12}{0.02}$$

$$= 256\ steps$$

Number of bits used, $N = log_2 256$

$= 8$ bits

(b) $V_{in} = \frac{m}{2^N} V_{REF}$

$$m = \frac{V_{in}}{V_{REF}} 2^N$$

$$= \frac{3}{5.12} 256$$

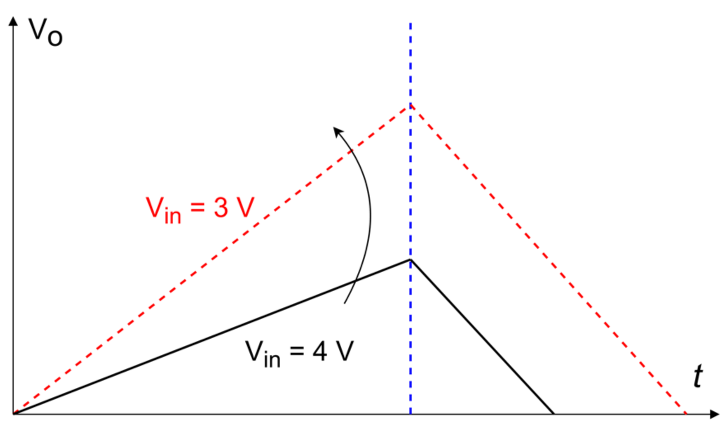
$$= 150\ counts$$

Time required, $t = t_1 + t_2$

$$= \frac{256}{2 \times 10^6} + \frac{150}{2 \times 10^6}$$

$$= 2.03 \times 10^{-4}\ seconds$$

(c)



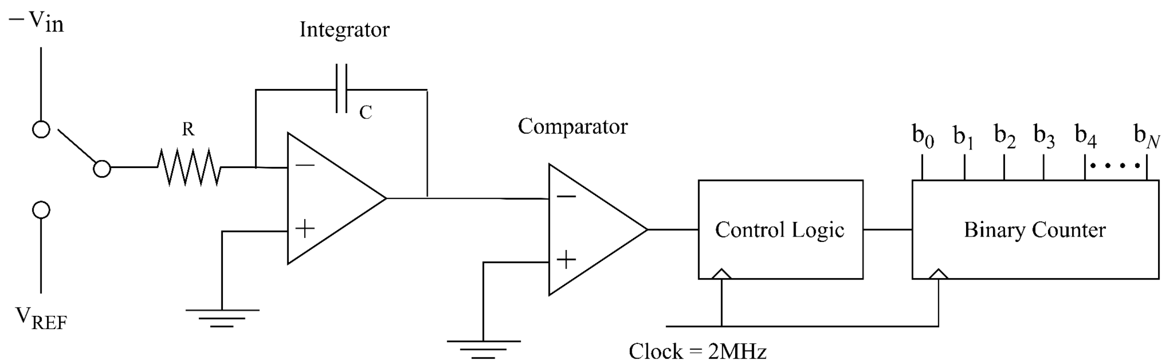
The rising portion of the graph has a slope of $\frac{V_{in}}{RC}$

When $V_{in} = 3$ V, slope is $\frac{3}{RC}$

When input increases to $V_{in} = 4$ V, slope is $\frac{4}{RC}$. The slope increases/ becomes steeper

For both inputs, falling portion of the graphs have a slope of $-\frac{V_{REF}}{RC}$

Solution - Set B



A dual slope A/D converter has V_{REF} = 7.68 V, and a N bit counter that outputs N-bit representation of the input signal. A 2 MHz clock is used for the clock and counter circuits

CO2	**(a)**	**Determine** the input voltage range. If step size is 0.03 V, **determine** the total number of steps and the number of bits used in this ADC.	**7**
CO2	**(b)**	For input voltage, V_{in} = 6 V, **determine** the ADC output count. **Calculate** the time required to get this output.	**7**
CO2	**(c)**	**Draw Vo vs time graph** for the input voltage V_{in} mentioned in (b). If V_{in} = 4V, how the graph will change? Explain briefly	**6**

Solution:

(a) Input Voltage range = (V_{REF} - 0) = 7.68 V

$$\text{Total Number of Steps} = \frac{Voltage\ range}{step\ size}$$
$$= \frac{7.68}{0.03}$$
$$= 256\ steps$$

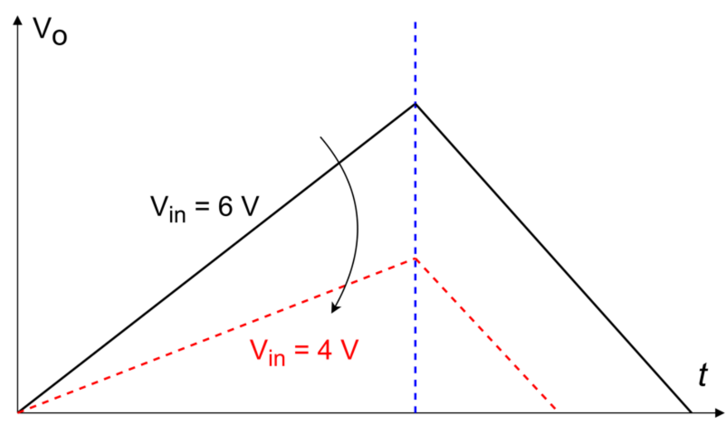
Number of bits used, $N = log_2 256$

$= 8$ bits

(b) $$V_{in} = \frac{m}{2^N} V_{REF}$$
$$m = \frac{V_{in}}{V_{REF}} 2^N$$
$$= \frac{6}{7.68} 256$$
$$= 200\ counts$$

$$\text{Time required, } t = t_1 + t_2$$
$$= \frac{256}{2 \times 10^6} + \frac{200}{2 \times 10^6}$$
$$= 2.28 \times 10^{-4}\ seconds$$

(c)



The rising portion of the graph has a slope of $\frac{V_{in}}{RC}$

When V_{in} = 6 V, slope is $\frac{6}{RC}$

When input decreases to V_{in} = 4 V, slope becomes $\frac{4}{RC}$. The slope decreases/ becomes flatter

For both inputs, falling portion of the graphs have a slope of $-\frac{V_{REF}}{RC}$

(TTL, MOSFET)

Set-A (Q4)

(a)

$$I_1 = \frac{6-2.3}{10} = \boxed{0.37\,mA}$$

$$I_{EX} = I_{EY} = I_{EZ} = \beta_R I_1 = 0.037\,mA$$

$$I_{B2} = I_1 + 3I_{EX} = \boxed{0.481\,mA}$$

$$I_2 = \frac{6-0.9}{4} = \boxed{1.275\,mA}$$

$$I_4 = \frac{0.8}{2} = \boxed{0.4\,mA}$$

$$I_{B0} = I_2 + I_{B2} - I_4 = \boxed{1.356\,mA}$$

$$I_3 = \frac{6-0.1}{5} = \boxed{1.18\,mA}$$

(b) I_1, I_2, I_3, I_4 same.

$$I_{B2} = I_1 + 2I_{Ex} = 0.444\,mA$$

$$I_{B0} = 1.275 + 0.444 - 0.4 = 1.319\,mA$$

$$I_3/I_{B0} = 0.87 < \beta_F$$

$\therefore$ $\boxed{\text{Saturation}}$

Set-B (Q3)

(a)

$$I_1 = \frac{5-2.3}{10} = \boxed{0.27\,mA}$$

$$I_{EX} = I_{EY} = I_{EZ} = \beta_R I_1 = 0.054\,mA$$

$$I_{B2} = I_1 + 3I_{EX} = \boxed{0.432\,mA}$$

$$I_2 = \frac{5-1}{4} = \boxed{1\,mA}$$

$$I_4 = \frac{0.8}{2} = \boxed{0.4\,mA}$$

$$I_{B0} = I_2 + I_{B2} - I_4 = \boxed{1.032\,mA}$$

$$I_3 = \frac{5-0.2}{5} = \boxed{0.96\,mA}$$

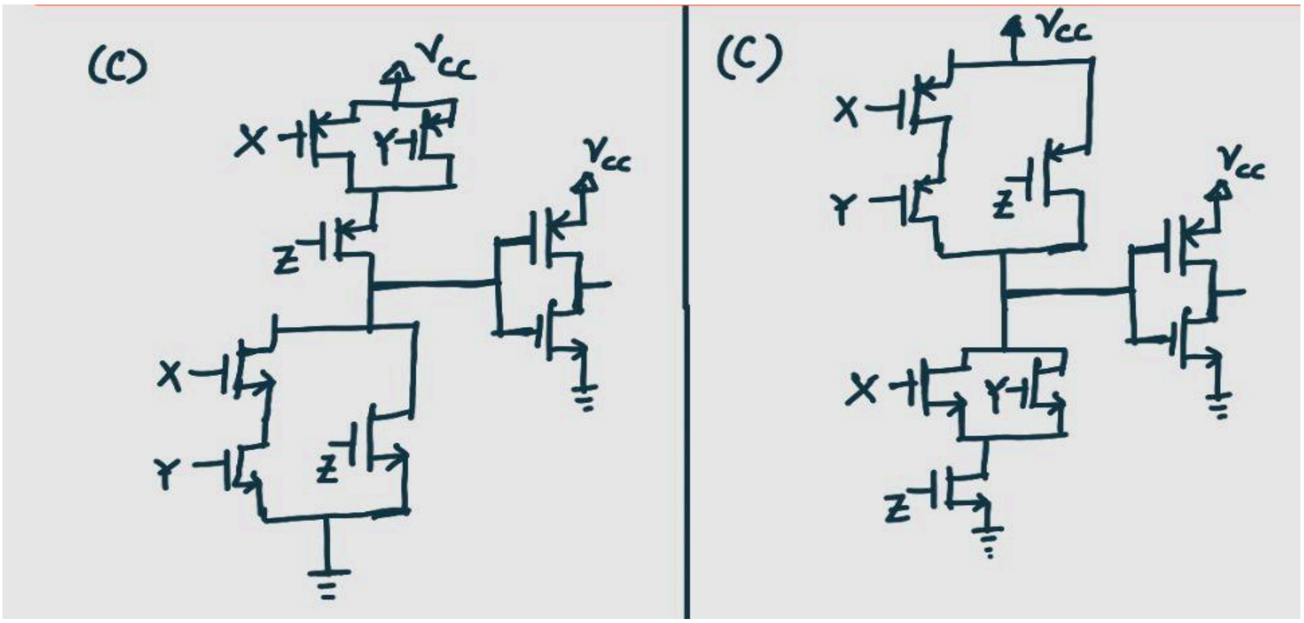
(b) I_1, I_2, I_3, I_4 same.

$$I_{B2} = I_1 + 2I_{Ex} = 0.378\,mA$$

$$I_{B0} = 1 + 0.378 - 0.4 = 0.978\,mA$$

$$I_3/I_{B0} = 0.982 < \beta_F$$

$\therefore$ $\boxed{\text{Saturation}}$


(C)
Vcc
X
Y
Z
Vcc
X
Y
Z
(C)
Vcc
X
Y
Z
Vcc
X
Y
Z

ECL

Solution- Set A

Set A   Q2

(a) V_{o1} Low $= 2.5V$ $V_{o2} = 5 - 0.7 = 4.3V$ (high)

$$I_{CR} = \frac{5 - V_{B4}}{0.75K} \qquad V_{B4} = 2.5 + 0.7 = 3.2V$$

$$I_{CR} = 2.4mA \qquad \boxed{I_E = I_{CR} = 2.4mA}$$

(b) $V_{o2} = 2.8V$

$$V_{B3} = 2.8 + 0.7 = 3.5V$$

$$V_{C1} = V_{B3} = 3.5V$$

$$\frac{5 - 3.5}{R_{C1}} = I_E = 2.4mA$$

$$\boxed{R_{C1\,max} = 0.625}$$

allow

$$\frac{5 - X}{R_{C1\,(not\ max)}} = 2.4mA$$

$$2.8 \leq X < 4$$

but cut marks as $R_c \neq R_{cmax}$

(c) $V_{ref} = \dfrac{4.3 + 2.5}{2}$ or $V_{ref} = \dfrac{4.3 + 2.8}{2}$

$\boxed{= 3.4V}$ or $\boxed{= 3.55V}$

Solution- Set B

Set B Q1

(a) $V_{o1} = 3V$ $V_{B4} = 3.7V$ $V_{o2} = 6 - 0.7 = 5.3V$

$$I_{CR} = \frac{6 - 3.7}{0.7} = 3.29\,mA$$

$$\boxed{I_E = I_{CR} = 3.29\,mA}$$

(b) $V_{o2} = 2.8\,V$

$V_{B3} = 2.8 + 0.7 = 3.5V$

$V_{C1} = V_{B3}$

$$\frac{6 - 3.5}{R_{C1}} = 3.29 = I_E$$

$$\boxed{R_{C1\,max} = 0.760\,k\Omega}$$

allow

$$\frac{5 - X}{R_{C1\,(not\ max)}} = 2.4\,mA$$

$2.8 \leq X < 4$

but cut marks as $R_C \neq R_{Cmax}$

(c) $$V_{ref} = \frac{2.8 + 5.3}{2}$$ or $$= \frac{3 + 5.3}{2}$$

$\boxed{= 4.05\,V}$ or $\boxed{= 4.15\,V}$